

Do the Functional Brain Networks of Attention-Deficit/Hyperactivity Disorder (ADHD) Children Show Altered Small-World Properties?

Abstract

Attention-Deficit/Hyperactivity Disorder (ADHD) is the most commonly diagnosed childhood neurodevelopmental disorder, increasingly viewed as a network disorder rather than a focal brain dysfunction. In this study, we investigated whether ADHD is associated with a functional network topology that shifts toward a more regular-like organization. Using graph theory, the networks were constructed at six proportional thresholds, and metrics including path length, clustering coefficient, global efficiency, local efficiency, and small-worldness were computed. Our results showed that ADHD involves functional network reorganization toward a more random topology rather than a complete architectural disruption, with ADHD patients demonstrating significantly shorter characteristic path lengths across all thresholds and reduced clustering at higher densities after mean connectivity control. Global efficiency was elevated at sparse densities, while local efficiency showed no differences. These findings suggest that the disorder reflects network reconfiguration rather than network breakdown.

1.0 Background

The most commonly diagnosed neurodevelopmental disorder in childhood is Attention-Deficit/Hyperactivity Disorder (ADHD)¹. Previous studies show that, besides the behavioral disturbances exhibited by ADHD children, children with ADHD also exhibit functional and structural abnormalities in distributed neural systems, mainly linked with cognitive control, attention, and motivation/reward functions.² Increasingly, ADHD alongside many other psychopathologies, is being viewed not only as an anomaly of the focal brain region, but as disorders arising from dysfunction of distributed brain networks³.

Network neuroscience provides a framework for characterizing large-scale brain organization by modeling the brain as a graph composed of nodes and edges⁶. The development of noninvasive neurophysiological and neuroimaging techniques as well as graph network analysis tools allows to model the brain as a complex network and to further explore topological organization of resultant networks.⁷ Using these approaches, researchers have consistently found that the human brain functional and structural networks exhibit many nontrivial topological properties such as small-worldness, modularity, and highly connected hub regions⁸. Several recent studies have utilized these connectome-based approaches to demonstrate topological organization disruption in both functional and structural brain networks in children with ADHD⁴, providing novel insights into the core pathophysiological mechanism of network dysfunction.

In this study, we applied graph theoretical analyses to resting-state fMRI-derived functional connectivity networks to compare the complexity of brain networks between controls and ADHD patients. We hypothesized that ADHD networks would show altered small-world properties compared to healthy controls. Specifically, we predict that ADHD networks may shift toward a more regular-like topology, characterized by higher local efficiencies and lower global efficiencies

2.0 Methods

2.0.1 Description of the Dataset

Functional MRI connectivity matrices for ADHD and control participants were downloaded from the open data Kaggle Dataset as provided by the Healthy Brain Network (HBN) as part of the Women in Data Science (WiDS), 2025 hackathon. The connectivity dataset comprised 1,213 participants, each with 19,901 unique pairwise connections. Each row is a participant, each column represents a unique pairwise connection between brain regions, and the values are the Pearson correlation coefficients for each connection. The link to the dataset can be found:

<https://www.kaggle.com/competitions/widsdatathon2025/data>

2.0.2 Data Preparation

The fMRI connectome matrices dataset was merged with the clinical information data containing patients' IDs with ADHD outcomes. The merged dataset was split by the diagnostic outcome ("ADHD_Outcome") into patient (n=831) and control (n=382) groups. For each group, participant IDs and other labels were removed, retaining only the 19,901 connectivity features. The flattened connectivity vectors were then reconstructed into full 200x200 symmetric connectivity matrices for each participant. The reconstruction yielded two three-dimensional arrays: fmri_ADHD (200×200×831), containing connectivity matrices for all ADHD patients, and fmri_control (200×200×382), containing matrices for all healthy controls. These reconstructed matrices served as input for all the subsequent graph analyses.

2.0.3 Graph Analyses

Graph metrics were computed using the Brain Connectivity Toolbox (BCT) implemented in MATLAB. Networks were constructed by applying proportional thresholding at six different threshold levels (5%, 10%, 15%, 20%, 25%, and 30%). For each threshold, five graph metrics (Clustering Coefficient, Characteristic Path Length, Global Efficiency, Local Efficiency, and small worldness) were calculated.

2.0.4 Regression of Mean Functional Connectivity

To account for individual differences in overall functional connectivity strengths that could confound group comparisons, we regressed out the mean functional connectivity from each graph metric prior to statistical testing. For each participant, the mean functional connectivity was calculated as the average of all positive connections in the upper triangle of the unthresholded connectivity matrix. Within each group, linear regression was performed to model the relationship between each graph metric and the mean functional connectivity. Residuals from these regressions, representing variances in graph metrics not explained by overall connectivity strength, were retained and centered by adding back the group mean. We applied this procedure to all four graph metrics for both groups at the threshold level.

2.0.5 Statistical Analyses

Group differences in graph metrics were assessed using permutation testing with 10,000 permutations. Statistical analyses were conducted both with and without regressing mean functional connectivity (FC) to control for individual differences in overall connectivity strength. Two-tailed p-values were calculated by comparing observed group differences to null distributions generated by randomly shuffling group labels.

3.0 Results

Group differences between ADHD and control participants were assessed for characteristic path length, clustering coefficient, global efficiency, local efficiency, and small worldness across proportional thresholds ranging from 5% to 30%. Results are reported for analyses both with and without mean functional connectivity (FC) regression. After mean FC regression, ADHD patients showed significantly shorter characteristic path lengths than controls across all thresholds (all $p < 0.05$; Table 1). The clustering coefficient was significantly lower in ADHD at the 25% and 30% thresholds ($p < 0.05$; Table 2). Global efficiency was significantly higher in ADHD at the 5% and 10% thresholds ($p < 0.05$; Table 3). Local efficiency showed no significant group differences at any threshold (Table 4). Small-worldness index showed a significant difference only at the 30% threshold ($p = 0.0241$; Table 5). Without mean FC regression, global efficiency remained statistically significant at the 5% threshold ($p = 0.0089$), while other metrics showed non-significant group differences. The proportion of subjects exhibiting small-world properties ($\sigma > 1.0$) remained high in both groups across all thresholds.

Table 1. Characteristic path length comparison between ADHD and control groups with and without mean functional connectivity regression.

Threshold	Observed Difference	p-value (With FC Regression)	p-value (Without FC Regression)
5%	-0.0559	0.0319*	0.0653
10%	-0.0537	0.0002***	0.0098**
15%	-0.0363	0.0041**	0.0755
20%	-0.0293	0.0131*	0.1728
25%	-0.0274	0.0095**	0.2206
30%	-0.0266	0.0118*	0.2385

Note. Observed difference = ADHD mean - Control mean. Negative values indicate shorter path length in ADHD group. P-values obtained from non-parametric permutation testing with 10,000 permutations. FC = functional connectivity. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 2. Clustering coefficient comparison between ADHD and control groups with and without mean functional connectivity regression.

Threshold	Observed Difference	p-value (With FC Regression)	p-value (Without FC Regression)
5%	0.0007	0.6739	0.7028
10%	-0.0017	0.3191	0.4190
15%	-0.0024	0.1345	0.2761
20%	-0.0024	0.0781	0.2626
25%	-0.0023	0.0482*	0.2763
30%	-0.0022	0.0213*	0.2725

Table 3. Global efficiency comparison between ADHD and control groups with and without mean functional connectivity regression.

Threshold	Observed Difference	p-value (With FC Regression)	p-value (Without FC Regression)
5%	0.0019	0.0085**	0.0089**
10%	0.0011	0.0485*	0.1672
15%	0.0006	0.2495	0.5527
20%	0.0004	0.3968	0.7161
25%	0.0003	0.4403	0.7787
30%	0.0003	0.4684	0.7990

Table 4. Local efficiency comparison between ADHD and control groups with and without mean functional connectivity regression.

Threshold	Observed Difference	p-value (With FC Regression)	p-value (Without FC Regression)
5%	0.0027	0.1886	0.1959
10%	0.0004	0.8174	0.8560
15%	-0.0006	0.6553	0.7849
20%	-0.0010	0.3668	0.6474
25%	-0.0012	0.2057	0.5880

30%	-0.0013	0.0767	0.5414
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Table 5. Small-worldness index (σ) comparison between ADHD and control groups with and without mean functional connectivity regression.

Threshold	ADHD Mean (SD)	Control Mean (SD)	Observed Difference	p-value (With FC Regression)	p-value (Without FC Regression)
5%	3.187 (0.442)	3.153 (0.431)	0.0342	0.1065	0.2085
10%	2.378 (0.433)	2.373 (0.406)	0.0057	0.7358	0.8238
15%	1.868 (0.352)	1.881 (0.377)	-0.0123	0.3362	0.5810
20%	1.583 (0.330)	1.598 (0.352)	-0.0152	0.1300	0.4614
25%	1.372 (0.312)	1.386 (0.327)	-0.0139	0.0679	0.4820
30%	1.221 (0.293)	1.234 (0.307)	-0.0135	0.0241*	0.4661

4.0 Discussion

This study investigated functional brain network topology in ADHD patients compared to healthy controls. Both groups exhibited small-world architecture ($\sigma > 1$ in $>80\%$ of subjects), but ADHD patients showed significantly shorter characteristic path lengths across all thresholds and reduced clustering at higher network densities after controlling for mean functional connectivity. These findings suggest ADHD involves network reorganization toward a less modular, more randomly-configured topology rather than complete disruption of efficient architecture. Our hypothesis predicted a shift toward regular-like topology with increased clustering, longer paths, higher local efficiencies, and lower global efficiencies. The observed patterns does not support this, as they diverge from some literature. Wang et al. (2009) reported that children with ADHD showed higher local efficiency combined with lower global efficiency, suggesting a shift toward regular networks. In contrast, we found higher global efficiency at sparse thresholds (5-10%) with no significant local efficiency differences, suggesting enhanced global integration rather than increased segregation. These discrepancies may reflect developmental differences or methodological factors such as thresholding approaches and mean FC regression. We also tried to look at the effect of regressing out the mean functional connectivity. While group differences in path length reached significance after FC regression ($p < 0.05$ at all thresholds) but not without it, the observed differences were identical, and effect sizes were small. Therefore, we cannot conclude definitively that regression "improves" statistical power without considering practical significance. Future work should examine additional network properties, such as modularity, to directly quantify functional segregation, and compare alternative thresholding methods, including edge prevalence approaches. Overall, our findings demonstrate that ADHD involves reorganized rather than disrupted network architecture.

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